

Unit I	3 mark question and answer
	1. What is research? • Research is a systematic and organized process of investigating, studying, and analyzing information to discover new facts, verify existing knowledge, or solve specific problems. <u>Types of research</u> • Descriptive • Analytical • Applied • Fundamental • Quantitative • Qualitative
	2. Explain the objectives of research • The objectives of research are the specific goals or purposes. • It guide a researcher in conducting a study. • They define what the researcher aims to achieve and why the study is being carried out. <u>Key points of objectives research:</u> 1. To find out the real facts 2. To achieve the new thoughts 3. To evaluate the information 4. To test a hypothesis 5. To design or implement the research 6. To improve the understanding
	3. Define "research problem" • A research problem is a specific issue, concern, or gap in knowledge. • The researcher aims to investigate and address through a systematic and scientific inquiry. • It is a clear and concise statement that defines the focus of the research study. <u>Types of Research Problems</u>

1.Descriptive Research Problem

2. Explanatory (or Causal)

3. Exploratory Research Problem

Characteristics of a Research Problem

- Specific
- Clear
- Researchable
- Significant

4. Describe the concept of qualitative research

- Qualitative research is a type of research.
- It is focuses on understanding human behavior, experiences, attitudes, and social phenomena.
- It is non-numerical data such as words, images, observations, or interviews.

Approach

- Exploratory and inductive approach.
- Focuses on gaining a
- Emphasizes context, meaning

Data Collection

- Methods
- Interviews
- Focus groups
- Observations
- Content analysis

5. What is the difference between pure research and applied research?

Topic	Pure research	Applied research
Definition	Pure research, also known as basic or	Applied research focuses on

	fundamental research.	solving a specific problem or addressing a practical issue.	
Goal	Advance knowledge	Solve a problem	
Focus	Understanding principles	Practical application	
Methodology	It use experimental or exploratory methods	It use many experiment Method	
Out come	Contribute to theoretical knowledge	Practical solution	

6. List the components of research work

Components of Research Work

- Research work typically consists of several key components that are essential for a comprehensive and well-structured study.
- Here are the main components of research work.

List of research work

- 1.Introduction
2. Literature Review
3. Methodology
4. Data Collection
5. Data Analysis
6. Results and Discussion
7. Conclusion and Recommendations
8. References

7. Explain what is meant by action research

- Action research is a type of research that involves actively participating in the research context to identify and solve problems.

- It is a collaborative and iterative process that aims to bring about positive change and improvement in a specific setting.

Applications of Action Research

1. Education
2. Healthcare
3. Organizational development

Steps in Action Research

1. Identify a problem
2. Plan
3. Act
4. Observe
5. Reflect

8. How does experimental research differ from survey research?

Topic	Experimental research	Survey research?
Definition	Experimental research and survey research are two distinct research used to collect and analyze data.	Survey research is a method of collecting data by asking people questions through questionnaires or interviews.
• Purpose	To establish cause-and-methodologies effect relationships.	To describe, explore, or understand attitudes, opinions, behaviors, or characteristics of a large population.
Method	Controlled experiments random assignment,	Uses samples, structured questions (closed/open-

•		control groups, and treatment groups.	ended), statistical analysis; mostly observational, not manipu look	
9. What is the significance of the research process? Advance Knowledge: • Research contributes to the existing body of knowledge, helping to understand complex issues and phenomena. Solve Problems: • Research identifies solutions to real-world problems, informing decision-making and policy development. Foster Innovation: • Research drives innovation, leading to new discoveries, products, and services. Critical Thinking: • The research process develops critical thinking, analytical, and problem-solving skills. Improve Practices: • Research helps to evaluate and refine practices, leading to improved outcomes and efficiency.				
10. What are the main objectives of conducting research? To explore and understand • Research aims to explore and understand a particular phenomenon, issue, or problem. To identify and analyze • Research identifies and analyzes patterns, trends, and relationships within the data. To explain and predict: • Research seeks to explain and predict outcomes, behaviors, or phenomena				

To evaluate and assess

- Research evaluates and assesses the effectiveness of interventions, programs, or policies.

To inform decision-making

- Research provides evidence-based information to inform decision-making in various contexts.

To contribute to knowledge

- Research contributes to the existing body of knowledge in a particular field or discipline.

11. Define descriptive research

- Descriptive research is a type of research methodology that aims to accurately describe a phenomenon, situation.
 - when the phenomenon occurred.
 - It involves collecting and analyzing data to provide a detailed description of the research subject.

Methods

1. Surveys
2. Interviews
3. Observations
4. Case studies

12. Explain exploratory research with an example

- Exploratory research is a type of research that aims to gain a better understanding of a topic, issue, or phenomenon.
 - It is used to explore a new area of study, identify patterns or trends, and generate hypotheses for further research.

Characteristics of Exploratory Research

- Flexible & Open-Ended
- Unstructured Methods
- Helps in Problem Definition

13. Describe the concept of research methodology

- Research is a systematic and organized process of investigating, studying.
- It is analyzing information to discover new facts, verify existing knowledge, or solve specific problems.

Types of Research

1. Qualitative research
2. Quantitative research
3. Mixed-methods research

Key Characteristics of Research

1. Systematic
2. Scientific
3. Objective
4. Methodical
5. Critical

14. What is survey research, and how is it used?

- Survey research is a method of collecting information from a group of people .
- It is one of the most common research methods in social sciences, business, marketing, education, and healthcare.
- Because it helps gather opinions, behaviors, experiences, or characteristics from a large population in a relatively short time.

Uses of Survey Research

1. Understanding Opinions
2. Market Research
3. Academic Research

15. Differentiate between research methods and research methodology

Aspect	Research Methods	Research Methodology
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Meaning	Used to collect and analyze data	The overall framework, strategy, and rationale
Focus	How to collect and analyze data	used and how the research process is structured.
Nature	Practical and procedural	Theoretical and philosophical.
Scope	only deals with the actual steps of conducting research	covers the principles, logic, and justification

16. What is the purpose of inferential research?

- Instead of studying the entire population researchers analyze a smaller group.
- It use statistical techniques to infer patterns, relationships.
- It apply to the whole population.

Main Purposes

- Generalization
- Hypothesis Testing
- Prediction

17. What is the simulation approach? Give an example.

- The simulation approach in research is a method.
- where a real-world system, process, or environment is imitated using.
- A model to study its behavior under different conditions.
- Instead of experimenting directly with the actual system researchers build a computer-based or mathematical model.
- It run simulations to observe outcomes.

18. What is the special significance of research in solving operational and planning problems of business and industry?

• Research plays a critical role in both the day-to-day operations (short-term issues) and long-term planning (strategic decisions) of business and industry.

Significance of research

- In Operational Problems
- In Planning Problems
- Special Significance

19. How can research methods be grouped into three categories?

• Research methods are usually grouped based on how data is collected, analyzed, and interpreted. A common way is to classify them into three broad categories.

three broad categories.

- Quantitative Research Methods
- Qualitative Research Methods
- Mixed Research Methods

20. What are the basic postulates of the scientific method?

- The scientific method is the foundation of all systematic research.
- It is built on certain basic postulates (assumptions/principles) that guide how knowledge is acquired, tested, and validated.

Basic

- Objectivity
- Verifiability
- Generalization

Unit II 3 marks question and answer

21. Define research design.

- A research design is the overall plan, structure, and strategy chosen.
- It integrates the different components of research in a coherent and logical way.
- It acts as a blueprint or roadmap that guides the researcher in the collection, measurement, and analysis of data.

Types of Research Design

- Exploratory Design

- Descriptive Design
- Experimental Design

Key Points

- Framework
- Guidance
- Minimizes Bias

22. What are the key elements of a research design?

1. Research Problem/Objective

- Clearly define what you want to study.
- The research objective guides the design and methods.

2. Hypothesis or Research Questions

- A testable statement predicting a relationship between variables.
- Open-ended questions the study aims to answer.

3. Type of Research

- Determines the overall approach.
- Understanding new problems.
- Describing characteristics

4. Time Frame

- A clear schedule for completing each stage of the research.

5. Sampling Design

- The full group you want to study.
- The subset you will actually study.

23. Explain the purpose of a literature review in research

- A literature review is the systematic study and analysis of existing research, books, articles.
- The other sources relevant to the research problem.
 - Its purpose is to provide a foundation and context for the study.

Key Purposes

1. Identifies Research Gaps

- Helps the researcher find areas that have not been studied or require further investigation.

2. Builds Theoretical Framework

- Provides concepts, theories, and models that guide the study.

3. Avoids Duplication

- Ensures the research adds original value rather than repeating earlier work.

4. Clarifies Concepts and Definitions

- Helps define key terms and variables clearly for the study.

5. Guides Methodology

- Shows which research methods have been effective or ineffective in past studies.

24. Describe the difference between inductive and deductive reasoning

Aspect	Inductive Reasoning	Deductive Reasoning
Direction	Specific → General	General → Specific
Approach	Bottom-up	Top-down
Purpose	To develop new theories/hypotheses	To test existing theories/hypotheses
Certainty	Conclusions are probable	Conclusions are certain (if premises are true)
Example	Observing many students who exercise are less stressed	Conclude exercise reduces stress

25. What are common sources of research problems?

1. Personal Experiences

- Everyday life situations often highlight issues worth studying.

Example: A student struggles with time management → leads to a study on productivity tools.

2. Practical Issues in Business

- Real-world challenges often generate research problems.

Example: Declining sales → research on consumer buying behavior.

3. Policy and Government Reports

- Government surveys, statistics, and policies often highlight problem areas.

Example: Census data showing high unemployment → research on job market trends

26. How is a research problem formulated?

- Formulating a research problem is the first and most critical step in the research process, because a well-defined problem guides the entire study.

Steps in Formulating a Research Problem

1. Identify a Broad Area of Interest

- Begin with a general subject area that interests you.

2. Review Existing Literature

- Study previous research to understand what is already known.

3. Define the Variables

- Clearly identify the dependent and independent variables.

27. What is a null hypothesis?

- Null Hypothesis (H_0)
- A null hypothesis is a type of hypothesis in research that states there is no significant relationship, effect, or difference between the variables being studied.
- It assumes that any observed difference in data is due to chance or random variation, not because of a true effect.

Purpose of a Null Hypothesis

1. Provides a baseline for statistical testing.
2. Helps researchers use data to accept or reject the claim.
3. Ensures conclusions are based on evidence, not assumptions.

28. Define a critical literature review

- A critical literature review is a detailed, evaluative, and analytical.
- It study of existing research and publications on a particular topic.
- A simple summary, it does not just describe what other researchers have said — it examines, compares, critiques, and synthesizes.

- The existing knowledge to highlight strengths, weaknesses, gaps, and contradictions.
- A critical literature review goes beyond “what is known” to ask “how reliable is this knowledge, what is missing, and what needs to be done next?”

29. What is a research paradigm?

- A research paradigm is a philosophical framework or set of beliefs that guides how research should be conducted.

It reflects a researcher’s view about:

1. Reality (Ontology) → What is the nature of reality? Does it exist independently, or is it socially constructed?

2. Knowledge (Epistemology) → How can we know reality? What counts as valid knowledge?

3. Methods (Methodology) → What strategies and tools should we use to study reality.

Main Types of Research Paradigms

1. Positivism
2. Interpretivism
3. Critical Paradigm
4. Pragmatism

30. Explain the components of a good research design

Components of a Good Research Design

1. Clear Research Problem and Objectives

- The design should begin with a well-defined problem and specific objectives.
- This gives direction and scope to the study.

2. Hypothesis or Research Questions

- States the expected relationship between variables (hypothesis) or poses clear research questions.

3. Variables and Their Definitions

- Independent variables (cause/factor being studied).
- Dependent variables (outcome/result).

- Control variables (kept constant).

31. How does one develop a hypothesis in research?

Steps in Developing a Hypothesis

1. Identify the Research Problem

- Start with a specific issue, question, or gap in knowledge.

2. Review the Literature

- Study existing theories, previous findings, and relevant concepts.
- This helps in identifying variables and possible relationships.

3. Define Variables

- Break down the problem into measurable elements:
- Independent variable (cause) – e.g., "study hours"
- Dependent variable (effect) – e.g., "academic performance"

4. Formulate Relationships

- Decide on the expected connection between the variables.

5. Frame the Hypothesis Clearly

- State it in a precise, testable, and measurable

32. What is the purpose of hypothesis testing?

Key Purposes:

1. To Test Assumptions

- It helps verify whether the assumption (hypothesis) made about a population is true or not.

2. To Make Decisions Based on Evidence

- Instead of relying on guesswork, hypothesis testing uses statistical evidence to accept or reject a claim.

3. To Validate Research Findings

- It ensures that the observed results are not due to chance but represent a real effect or relationship.

4. To Compare Groups or Variables

- Testing whether two teaching methods produce different student performance levels.

5. To Minimize Errors in Conclusions

- Hypothesis testing uses probability to control for Type I error (false positive) and Type II error (false negative).

33. Describe the significance of a literature review in developing research objectives

1. Identifies Research Gaps

- By reviewing previous studies, the researcher can spot what has not been studied or areas needing further exploration.
- This directly helps in framing specific and relevant objectives.

2. Provides Theoretical Foundation

- It offers concepts, theories, and models that guide the formulation of objectives.

3. Avoids Duplication

- Ensures that the research objectives are not a mere repetition of previous studies but add original value.

4. Clarifies Variables and Relationships

- Literature helps identify independent and dependent variables and possible cause–effect links, which are then translated into objectives.

5. Refines the Scope of Research

- Prevents overly broad or vague objectives by highlighting feasible and specific areas to focus on.

34. What are the features of a good hypothesis?

Features of a Good Hypothesis

1. Clear and Precise

- The hypothesis should be simple, specific, and unambiguous.

2. Testable

- It must be possible to verify the hypothesis using empirical data through observation, experiments, or surveys.

3. Based on Existing Knowledge

- A good hypothesis is grounded in theory, previous studies, or logical reasoning.

4. Specific and Limited in Scope

- It should focus on a particular problem and clearly define variables and relationships.

5. Consistent with Available Techniques

- The hypothesis should be testable with available tools, methods, and resources.

35. What is the necessity of defining a problem?

1. Provides Clear Direction

- A properly defined problem sets the scope and boundaries of the research.

2. Determines Objectives

- Research objectives can only be formulated when the problem is clearly understood.

3. Guides Literature Review & Hypothesis

- Problem definition helps in selecting relevant literature and developing testable hypotheses.

4. Helps in Choosing Methodology

- The research design, data collection methods, and analysis techniques depend on the nature of the problem.

5. Avoids Wastage of Resources

- Saves time, money, and effort by preventing unnecessary or irrelevant investigations.

36. What does rephrasing the research problem mean?

- Rephrasing the research problem means restating the original research problem in a more precise, specific, and researchable form.
- It is the process of converting a broad, vague, or general problem into a clear, well-defined problem statement that can guide the formulation of objectives, hypotheses, and methodology.

Why Rephrasing is Needed

1. To Improve Clarity
2. To Make It Researchable
3. To Focus the Scope

4. 4. To Define Variable

37. How can the overall research design be split into four parts?

- The research design is the blueprint of research, guiding how data will be collected, analyzed, and interpreted.

It can be split into four main parts:

1. Sampling Design –

- Who is to be studied.

2. Observational/Data Collection Design

- How data is gathered.

3. Statistical Design –

- How data is analyzed.

4. Operational Design –

- How the research

38. What are the factors involved in research design for a specific problem?

1. Nature of the Research Problem

- Whether the problem is exploratory, descriptive, diagnostic, or experimental determines the type of design to be used.

2. Objectives of the Study

- The purpose of research (to explore, describe, test relationships, or establish causality) influences the choice of methods.

3. Types and Sources of Data

- Whether the study needs primary data (collected firsthand) or secondary data (existing data) affects the design.

4. Sampling Design

- Selection of population, sample size, and sampling technique (random, stratified, cluster, etc.) is crucial.

5. Methods of Data Collection

- Choice between survey, interview, observation, experiment, or case study depends on the problem.

39. Define experience survey.

- An experience survey is a method of exploratory research.
- which the researcher collects information about a problem by seeking the opinions, experiences.
- who are knowledgeable, experienced, or experts in the field.
- It helps in understanding the nature of the problem, identifying variables, and generating hypotheses for further study.

Key Points

- It is qualitative in nature.
- Involves interviews or discussions with experts, practitioners, or people who have direct experience.
- Used in the early stages of research to clarify concepts and refine the research problem.
- Provides ideas, new perspectives, and practical insights.

40.What is a hypothesis, and why is it important?

- A hypothesis is a tentative statement or proposition about the relationship between two or more variables that can be tested empirically.
- It predicts an expected outcome based on theory, prior research, or observation.

Importance of a Hypothesis

1. Guides the Research
2. Helps in Defining Variables
3. Facilitates Data Collection
4. Forms the Basis for Analysis
5. Supports Scientific Inquiry